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Gina Cody School of Engineering and Computer Science
Department of Mechanical, Industrial and Aerospace Engineering

INDU 211: INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

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THE FUTURE OF INDUSTRIAL ENGINEERING IN THE GENERATIVE AI ERA:

Opportunities, Limitations, and the Critical Role of Human Engineers

Final Term Paper Project Report

Prepared By: Team 4 — Production and Systems Engineering Group

Student Name	Student ID	Primary Area of Contribution
Student 1 (Team Lead & Coordinator)	40XXXXXX	Systems Engineering Framework, Introduction & Report Integration
Student 2 (Technology Researcher)	40XXXXXX	AI vs. Generative AI Concepts and Workplace Transformation
Student 3 (Operations Analyst)	40XXXXXX	Automation Potential Across Core Industrial Engineering Areas
Student 4 (Risk & Ethics Analyst)	40XXXXXX	AI Limitations, Technical Hallucinations, and OIQ Legal Ethics
Student 5 (Human Factors Specialist)	40XXXXXX	Human-in-the-Loop Concept and Future Student Competencies

Instructor & Evaluation Committee:

Department of Mechanical, Industrial and Aerospace Engineering
Concordia University, Montreal, Quebec, Canada

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EXECUTIVE SUMMARY

Artificial Intelligence (AI) and Generative AI (GenAI) tools like ChatGPT, automated scheduling software, and computer vision systems are rapidly entering modern factories and engineering offices. This term paper investigates what these new technologies mean for the field of Industrial Engineering (IE).

Rather than treating AI as an abstract computer science subject, our report builds directly on the foundational ideas taught in INDU 211: the **Conversion Process Model** (turning raw inputs into useful outputs), **Open-Loop versus Closed-Loop feedback systems**, and the three levels of manufacturing decisions (**Strategic, Tactical, and Operational Control**). We examine six main areas of industrial engineering: production scheduling, factory layout planning, assembly line balancing, inventory management, quality control, and workplace ergonomics. For each area, we explain in clear and practical terms what AI can handle automatically and why human engineers are still needed.

Our findings show that while AI is great at crunching large amounts of data, finding patterns, and generating preliminary reports or schedules, it has serious weaknesses. AI models frequently "hallucinate" incorrect numbers, have no physical common sense about how machines work in the real world, operate as unexplainable "black boxes," and cannot be held legally responsible for worker safety under Canadian engineering law and the *Ordre des ingénieurs du Québec* (OIQ).

We conclude that the future of manufacturing relies on a **Human-in-the-Loop** approach: AI works as a fast digital assistant, while human engineers provide practical problem-solving, common sense, safety checks, leadership, and ethical responsibility. Finally, we provide a clear guide to the basic technical and communication skills that new engineering students should develop to succeed in an AI-assisted workplace.

Keywords: Industrial Engineering, Generative AI, Production Systems, Closed-Loop Control, Factory Layout, Assembly Lines, Quality Control, Human-in-the-Loop, Engineering Ethics.

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1.0 Introduction: What is Industrial Engineering and What is Generative AI?

1.1 The Core Purpose of Industrial Engineering

Industrial Engineering (IE) is an engineering branch that focuses on making entire systems and processes work better, faster, safer, and more cost-effectively. While mechanical engineers design individual machine parts and civil engineers build roads and bridges, industrial engineers look at the whole picture. We organize and improve the complete system that connects people, machinery, materials, energy, and information.

The word *engineering* comes from the Latin root *ingenium*, which means natural talent, clever invention, or human ingenuity. Science focuses on discovering knowledge and understanding how nature works through experiments. Engineering, on the other hand, is about **synthesis**—taking that scientific knowledge and combining it with practical tools to solve real problems for human society. In a factory, industrial engineers design practical processes that minimize waste, keep production running on time, protect worker health, and maintain high product quality.

1.2 The Conversion Process Model: Inputs, Transformation, and Outputs

To understand any manufacturing plant or business, industrial engineers use the **Conversion Process Model** introduced in INDU 211. Every production system is composed of four main elements:

1. **Inputs:** The starting resources required to run the operation. These include raw materials (like sheet metal or plastic pellets), human labor, capital investment, operating instructions, machines, tooling, and electrical power.
2. **Transformation Process:** The physical or organizational steps that turn raw materials into useful goods. This includes processes like casting molten metal into molds, cutting parts on lathes and milling machines, welding components together, and packing finished items into boxes.
3. **Outputs:** The final results of the process. These include finished products delivered to customers, useful services, and unavoidable waste streams such as scrap metal chips, wasted heat, and idle waiting time.
4. **Feedback and Control:** Measuring tools, quality inspections, and production reports that track how well the plant is doing. This feedback tells managers and engineers whether they need to adjust machine settings or change schedules.

The main job of the industrial engineer is to make this conversion process as efficient as possible by removing unnecessary waste (*Muda* in Lean manufacturing) and ensuring smooth workflow.

1.3 Open-Loop vs. Closed-Loop Systems in Factory Operations

An important concept in systems engineering is the difference between open-loop and closed-loop control:

- **Open-Loop System:** A system that follows a pre-set command without checking what actually happens to the output. A simple example is a regular household toaster with a mechanical timer; it toasts the bread for two minutes regardless of whether the bread is frozen, fresh, or already burning.
- **Closed-Loop System:** A system that continuously monitors its actual output, compares the result to the desired goal, and makes adjustments to correct any error. A common example is a home heating thermostat; it measures room temperature and turns the furnace on or off to maintain the target comfort level.

Many computer algorithms and generative AI tools operate like open-loop systems. They produce an initial plan or schedule based on textbook assumptions. But a real factory is not a closed computer simulation. Tools break, raw materials arrive late, and machines leak oil. An industrial engineer uses closed-loop thinking: observing what is actually happening on the shop floor, gathering real-time feedback, and modifying the process so the factory stays on track.

1.4 The Three Levels of Engineering Decisions: Strategic, Tactical, and Control

Industrial engineers make decisions across three distinct timeframes:

- **Strategic Decisions (Long-Term: 3 to 10 Years):** High-level planning decisions made by plant executives and senior engineers. These include choosing what new products to manufacture, deciding where to build a new factory using facility location methods, and investing millions of dollars in automated production equipment.
- **Tactical Decisions (Medium-Term: Monthly to Quarterly):** Operational planning decisions that balance supply and demand. These include determining monthly production targets, deciding how many parts to keep in the warehouse, setting staffing schedules, and selecting raw material suppliers.
- **Operational Control Decisions (Short-Term: Daily to Hourly):** Immediate, hands-on decisions made right on the factory floor. These include assigning daily tasks to specific CNC machines, solving a sudden bottle-jam on an assembly line,

inspecting bad parts, and replacing broken cutting tools.

Generative AI tools are very helpful for operational calculations and tactical document drafting. However, high-stakes strategic decisions still require human judgment, business experience, and ethical vision.

2.0 Understanding the Technology: Traditional AI vs. Generative AI

To understand how artificial intelligence impacts industrial engineering, we first need to clarify the difference between traditional AI and Generative AI. While both rely on computer data, they serve very different purposes.

2.1 Traditional AI: Pattern Recognition and Predictive Maintenance

Traditional Artificial Intelligence and Machine Learning focus on **analyzing numbers, recognizing patterns, and making predictions**. These systems look at historical numerical data to answer specific questions:

- **Predictive Maintenance:** Sensors attached to a large stamping press record vibrations and temperatures every second. A machine learning program learns what normal vibrations feel like compared to the vibrations of a worn bearing. When it spots an abnormal vibration pattern, it alerts maintenance technicians to replace the bearing before the machine breaks down.
- **Demand Forecasting:** Algorithms examine past sales records, seasonal shopping trends, and weather data to estimate how many units customers will buy next month, helping engineers plan material purchases.
- **Camera-Based Defect Inspection:** Automated optical cameras on a conveyor belt scan finished components, classifying each part as either good or defective based on visual patterns learned from thousands of sample photos.

Traditional AI acts like a high-speed numerical calculator that identifies statistical trends in past records.

2.2 Generative AI: Creating Text, Code, and Design Concepts

Generative Artificial Intelligence (GenAI) is a newer development. Instead of simply sorting or predicting numbers, GenAI **creates new content** such as text, computer code, summaries, and 2D design layouts. These tools (like ChatGPT, Claude, and specialized industrial assistants) are trained on millions of technical books, operating manuals, and code samples.

In an industrial setting, Generative AI can:

- **Draft Standard Operating Procedures (SOPs):** A plant supervisor can type rough bullet points about how to safely operate a band saw, and the AI will expand those notes into a complete, professional step-by-step safety manual in seconds.
- **Write Data Analysis Scripts:** An engineer can ask the AI to write a short Python script that takes weekly production logs from Excel and automatically creates graphs of machine downtime.
- **Summarize Technical Manuals:** Field technicians can ask questions about an unfamiliar machine (such as *"What is the correct oil pressure for hydraulic pump #2?"*), and the AI quickly pulls the exact answer from a 600-page manufacturer PDF.

2.3 Side-by-Side Comparison of Both Technologies

The table below highlights the practical differences between traditional predictive AI and Generative AI in manufacturing:

Characteristic	Traditional Predictive AI	Generative AI (GenAI)
Main Function	Finds patterns and predicts numbers	Creates new text, code, and design ideas
Typical Input Data	Structured numbers, sensor readings, spreadsheets	Natural language questions, text manuals, code
Main Output	Predictions, failure probabilities, defect flags	Written reports, safety checklists, Python code
Common Factory Use	Machine health monitoring and sales forecasting	Drafting shift reports, writing scripts, training guides
Main Strength	Extremely fast and consistent with numbers	Easy to interact with using everyday English
Main Risk	Needs large amounts of clean numerical data	Can make up believable but false facts (hallucinations)

2.4 How AI Tools Change Daily Workplace Tasks

When combined, these two technologies change day-to-day factory work in three key ways:

1. **Cutting Down Paperwork:** Industrial engineers often spend hours every week writing shift summaries, documenting safety meetings, and filling out equipment logs. GenAI can draft these documents quickly, giving engineers more time to work on the shop floor.
2. **Making Data Easy to Access:** In the past, getting answers from a factory database required writing database queries or building complicated spreadsheet formulas. Now, a supervisor can simply ask in plain language: *"Which assembly line*

produced the most scrap yesterday?" and receive an answer immediately.

3. **Helping New Workers Learn Faster:** On complex assembly lines, new operators can use digital tablets with AI assistants to get clear, step-by-step assembly guidance whenever they encounter an unfamiliar part.

3.0 Which Areas of Industrial Engineering Can Be Automated?

Industrial engineering covers many different plant activities. Some tasks involve pure math and data processing, which computers handle very well. Other tasks involve human psychology, safety, and physical machines, where human hands and practical common sense are indispensable.

3.1 Production Scheduling and Machine Assignment

In any busy factory, customer orders arrive with different delivery dates, different processing steps, and different machine requirements. Deciding which order goes to which machine, and in what order, is known as **production scheduling**.

- **What AI Does Well:** Scheduling is a mathematical puzzle. An algorithm can test thousands of order sequences in seconds to find an arrangement that minimizes idle machine waiting time and groups similar jobs together to reduce machine setup time.
- **Why Humans Are Still Needed:** Real factories rarely run exactly as planned. A cutting tool might chip, an operator might call in sick, or a key customer might call asking for an emergency rush delivery. An algorithm cannot understand the business importance of keeping that customer happy, nor can it physically inspect whether an older machine can handle extra workload today. A human engineer must step in, talk to supervisors, and make a sensible compromise.

3.2 Facility Layout and Factory Space Design

Facility layout planning involves arranging machines, storage racks, aisles, and break rooms inside a building so that materials travel the shortest possible distance from raw storage to final shipping.

- **What AI Does Well:** Given a list of departments and how much material moves between them, generative layout software can quickly suggest dozens of alternative floor plan drawings that minimize total forklift travel.
- **Why Humans Are Still Needed:** Computer algorithms only look at 2D drawings. They do not know that a heavy 20-ton stamping press cannot be placed directly over an underground water pipe, that overhead crane paths must remain clear, or that forklift drivers need enough room to turn safely around corners without hitting fire exits. An industrial engineer must walk the physical plant floor, measure ceiling heights, check structural pillars, and ensure the space is safe and comfortable for workers.

3.3 Assembly Line Balancing and Bottleneck Elimination

On an assembly line (such as an automotive plant or an electronics plant), a product moves past multiple workstations. Each worker performs a specific set of assembly steps. If Workstation 3 takes 90 seconds while all other stations take 50 seconds, Workstation 3 becomes a **bottleneck**, slowing down the entire factory.

- **What AI Does Well:** Grouping assembly tasks so that each workstation takes approximately the same amount of time is a mathematical problem that algorithms solve easily. AI tools can balance hundreds of tasks across stations in seconds while respecting the order in which parts must be put together.
- **Why Humans Are Still Needed:** Computer models treat human workers like identical robots. In reality, human workers experience physical fatigue, some tasks require delicate finger dexterity, and different workers have different skill levels. If an algorithm assigns five exhausting, repetitive lifting tasks to the exact same workstation, the operator will experience muscle strain and start making mistakes. A human engineer understands worker fatigue and designs workstations that balance physical effort as well as time.

3.4 Inventory Management and Production Economics

Companies must keep enough raw materials on hand so production never stops, but storing too much inventory ties up expensive cash and warehouse space. Industrial engineers balance these costs by calculating how many parts to order at once (order quantities), when to reorder (reorder points), and how much extra buffer stock (safety stock) to keep in reserve.

- **What AI Does Well:** Inventory software tracks thousands of part numbers simultaneously. It monitors daily usage, updates safety stock levels automatically when demand changes, and sends electronic purchase orders to suppliers when stock gets low.
- **Why Humans Are Still Needed:** Inventory formulas assume that suppliers deliver on time and prices stay stable. In the real world, global shipping routes face winter storms, border delays, and sudden price spikes. An AI tool cannot pick up the phone

to negotiate prices with a local backup supplier, visit a vendor to check material quality, or decide when a company should buy extra steel before an expected tariff increase.

3.5 Quality Control and Defect Detection

Quality control ensures that manufactured products meet customer standards and that defective items are caught before leaving the building. In INDU 211, quality control emphasizes continuous improvement and waste reduction (*Kaizen* and Lean principles).

- **What AI Does Well:** High-speed digital cameras powered by AI can inspect hundreds of parts per minute on a fast-moving conveyor belt. They can spot micro-scratches on polished metal, detect missing screws, or check whether a label is crooked much faster than human eyes.
- **Why Humans Are Still Needed:** An automated camera can spot a bad part, but it cannot explain **why** the defect happened. Was the cutting tool worn down? Did the cooling fluid run out? Did a new supplier deliver contaminated plastic? Finding the root cause requires human investigation using diagnostic techniques like the **5 Whys** and the **Fishbone Diagram**. A human engineer inspects the machine, talks to the operator, and fixes the underlying physical problem.

3.6 Workplace Ergonomics and Worker Safety

Ergonomics (Human Factors) is the study of how people interact with their physical work environment. Good ergonomics prevents repetitive strain injuries, back pain, and long-term fatigue by designing benches, tools, and chairs to fit human bodies.

- **What AI Does Well:** Video cameras running AI pose-detection software can monitor shop-floor video feeds to flag unsafe lifting postures (such as bending with the back instead of the knees) or verify that workers are wearing hard hats and safety glasses in hazardous zones.
- **Why Humans Are Still Needed:** Ergonomics is about human empathy and comfort. A computer program cannot feel lower back pain or understand what it feels like to hold a vibrating pneumatic tool for eight hours. Continuous camera surveillance can also make workers feel stressed and distrustful. An industrial engineer talks directly with operators, listens to their physical complaints, and redesigns tools, workbench heights, and foot mats so workers can do their jobs safely and comfortably.

3.7 Summary Automation Matrix: What AI Can and Cannot Do

The following table provides a clear summary of how automation applies across all six core areas of industrial engineering:

Industrial Engineering Area	Potential for AI Automation	What AI Handles Automatically	Why Human Engineers Are Essential
1. Production Scheduling	High	Tests thousands of job sequences to minimize machine downtime and setup delays	Handles machine breakdowns, missing parts, and urgent customer priorities
2. Facility Layout Design	Medium	Generates alternative floor plans that reduce total travel distances for materials	Checks real physical limits: building pillars, ceiling height, fire exits, and forklift space
3. Assembly Line Balancing	Medium to High	Divides assembly steps between stations to minimize waiting time and bottlenecks	Prevents worker fatigue, balances physical strain, and accounts for different operator skill levels
4. Inventory Management	High	Tracks thousands of parts, predicts demand, and automatically reorders low stock	Negotiates with suppliers, manages shipping delays, and handles raw material shortages
5. Quality Control	High	Inspects parts at high speed with cameras to spot surface defects and missing pieces	Investigates the physical root causes of defects and fixes machine or tooling problems
6. Workplace Ergonomics	Low to Medium	Uses cameras to check safety gear compliance and detect awkward lifting postures	Listens to worker comfort complaints, redesigns physical tools, and builds a supportive safety culture

4.0 Common Errors, Risks, and Limitations of AI Platforms

While artificial intelligence is a valuable engineering aid, in real factory environments it exhibits serious flaws that can lead to expensive mistakes and safety hazards if not carefully supervised.

4.1 The Hallucination Problem: Believable but False Information

The most dangerous flaw of Large Language Models is known as **hallucination**. Because language models predict the next most likely word based on statistics rather than consulting a verified physical law, they can generate completely incorrect technical statements while sounding confident, polite, and authoritative.

- **Real-World Factory Scenario:** An engineer asks an AI assistant for the maximum safe operating temperature of a rubber

from a factory scenario. An engineer asks an AI assistant for the maximum safe operating temperature of a rubber conveyor belt. The AI replies confidently: *"This rubber is rated for continuous use up to 260°C."* In reality, that specific rubber begins to melt and release toxic fumes at 150°C. If the engineer does not verify the number in the manufacturer's official technical datasheet, the belt could catch fire during a night shift. In engineering, being "mostly right" is not good enough; an unverified decimal place or incorrect tolerance can cause severe damage.

4.2 Lack of Physical Common Sense and Real-World Context

AI models operate inside computer servers; they do not live in the physical world. They do not have an instinctive understanding of gravity, weight, heat, friction, or spatial clearance unless those factors are explicitly coded as mathematical rules.

- **Real-World Factory Scenario:** An automated layout software might design an efficient workstation layout where two large milling machines sit right next to each other. On paper, it saves space. But in reality, the machines are placed so close together that maintenance technicians cannot open the side electrical panels to change filters, and forklift drivers cannot bring in bins of raw parts. An engineer notices these physical issues immediately just by walking through the room.

4.3 The "Black Box" Problem and Auditing in Regulated Industries

Many advanced AI and neural network models are **"black boxes"**. They take in lots of data and produce a recommendation, but they cannot explain step-by-step why they chose that specific answer.

- In regulated manufacturing industries—such as aerospace (Bombardier, Pratt & Whitney), medical devices, and defense—government safety agencies (Transport Canada, the FAA, Health Canada) require complete documentation of how every component is made. If an airplane bracket fails in flight, engineers cannot tell crash investigators: *"We changed the heat treatment recipe because the AI told us to."* Every engineering choice must be backed by transparent physical formulas, verified test data, and documented human reasoning.

4.4 Data Privacy, Security, and Protecting Trade Secrets

Manufacturing companies spend millions of dollars developing proprietary production techniques, specialized tool designs, and secret chemical recipes that give them an edge over competitors.

- When employees copy and paste internal blueprints, machine settings, or production costs into free, public AI chatbots, that sensitive data can be stored on third-party cloud servers and potentially used to train future public models. This creates serious risks of leaking valuable company trade secrets. In addition, connecting AI tools directly to factory control systems increases the risk of cyberattacks, where outside hackers could tamper with physical machine settings.

4.5 Legal Responsibility and Professional Ethics under the OIQ

In Canada, the practice of engineering is strictly regulated by law to protect public safety. In Quebec, the governing body is the **Ordre des ingénieurs du Québec (OIQ)** under the *Engineers Act*.

- **An AI Cannot Be Held Legally Responsible:** An AI algorithm cannot earn an engineering degree, cannot take an ethical oath, cannot hold a professional license (ing. / P.Eng.), and cannot be taken to court for criminal negligence.
- **The Human Engineer Carries All Legal Liability:** Under the law, only a licensed human Professional Engineer has the legal authority to approve and seal engineering plans. If an automated assembly line hurts a worker or a defective product causes an accident, the human engineer who signed off on the system is held legally and financially accountable. Because the engineer carries all the personal and legal risk, they must never blindly follow a computer's suggestion without thorough verification.

5.0 The Critical Role of the Human Industrial Engineer

Since computers can calculate faster and process more information than any person, what is the true purpose of the human industrial engineer in modern industry? The answer is that the engineer's role becomes **more important, not less important**, moving from routine repetitive calculations to high-level system leadership.

5.1 Real-World Problem Solving and Unexpected Factory Crises

Factory floors are unpredictable environments. Machines break down, supplier deliveries get stuck in bad weather, power outages happen, and raw material shipments arrive with minor surface flaws.

- Machine learning algorithms work well during normal, predictable conditions where lots of past data exists. But when an unexpected emergency occurs—such as a sudden international supply chain shutdown or a factory flood—AI models struggle because they have never seen that situation before. Human engineers excel at practical creativity: finding local alternative suppliers, modifying machine tooling on short notice, and finding safe workarounds to keep the plant operating.

5.2 The "Human-in-the-Loop" Working Model

The most effective way to run modern manufacturing operations is the **Human-in-the-Loop (HITL)** approach:

1. **The AI Acts as a High-Speed Assistant:** It crunches large datasets, tracks subtle quality variations on conveyor belts, and generates several draft schedules or layout ideas.
2. **The Human Engineer Acts as the Decision-Maker:** The engineer reviews the suggestions, checks them against practical factory constraints and safety rules, and selects the best option.
3. **The System Learns from Human Feedback:** When the engineer adjusts an AI proposal, the software notes the correction so it produces better suggestions in the future.

This partnership combines the rapid calculating speed of computers with the practical common sense and safety awareness of human engineers.

5.3 Systems Synthesis vs. Computer Calculation

Computers are great at **analysis**—breaking a single problem down into small numbers and calculating an answer. But real engineering requires **synthesis**—bringing together many different, competing goals into one balanced, working solution.

- An industrial engineer must balance production speed against machine wear, equipment cost against worker safety, and urgent shipping deadlines against long-term maintenance needs. Making these trade-offs requires human judgment, ethical standards, and a deep understanding of business goals—qualities that cannot be reduced to a simple computer algorithm.

5.4 Shop-Floor Leadership, Communication, and Working with People

At its foundation, industrial engineering is about people. You can design a mathematically perfect assembly line on a computer screen, but if the shop-floor operators find the workstations frustrating, confusing, or painful to work on, productivity will drop.

- A computer cannot walk down the factory aisle, shake hands with an experienced machinist, listen to their concerns about an awkward foot pedal, or build a friendly, trusting team environment. Industrial engineers spend much of their day talking with machine operators, maintenance staff, quality inspectors, and company managers. Empathy, teamwork, negotiation, and leadership are uniquely human skills that no software can replace.

6.0 Essential Skills for Future Industrial Engineering Graduates

For new engineering students starting their studies, the growth of AI is not something to worry about; it is a valuable career opportunity. Manufacturing companies are actively looking for young engineers who know how to use AI tools responsibly while applying solid engineering fundamentals. Undergraduates should focus on four essential skill areas:

6.1 Basic Programming and Practical Data Literacy

Future industrial engineers do not need to become software developers, but they should feel comfortable handling production data:

- **Python Basics:** Python is the most popular programming language for data analysis. Students should learn how to write simple scripts to clean up messy maintenance logs, create clear graphs of factory downtime, and organize spreadsheet data.
- **Database Basics (SQL):** Modern factories record thousands of sensor and order records every day. Understanding how to write simple queries to pull the data you need from a database makes you much more productive.

6.2 Prompt Writing and Fact-Checking AI Answers

Using AI tools effectively requires two closely related habits:

- **Writing Clear Prompts:** Instead of asking vague questions, engineers should learn to give AI tools clear background details, specific goals, and clear limits (for example: *"You are an industrial engineer. Review these 50 machine downtime logs and list the top three causes of motor overheating"*).
- **Always Fact-Checking the Output:** A good engineer treats AI like a helpful but inexperienced assistant: its work is fast, but you must always double-check the numbers, formulas, and units against trusted textbooks and manufacturer datasheets before using them in a real report.

6.3 Strong Foundations in Core Industrial Engineering Concepts

Understanding core engineering principles becomes **more critical** in an automated world, not less:

- If you do not understand the practical concepts behind production scheduling, assembly line balancing, inventory planning, and quality control, you will have no way of knowing whether an AI suggestion is smart or disastrous.

- You need a strong personal understanding of engineering concepts to be a reliable judge of computer recommendations.

6.4 Teamwork, Communication, and Professional Responsibility

As repetitive calculation tasks become automated, companies value engineers primarily for their personal and communication strengths:

- **Clear Explanations:** The ability to explain complex technical data simply and clearly to both factory operators and company managers.
- **Good Teamwork:** Working well with people from different disciplines, including mechanical, electrical, and software team members.
- **Professional Ethics:** Remembering your primary duty as an engineer under the OIQ Code of Ethics—ensuring that worker health, public safety, and environmental protection always come before production speed or cost savings.

7.0 Conclusion

Artificial Intelligence and Generative AI represent a powerful new set of tools for industrial engineering, continuing a long history of manufacturing innovation that includes the assembly line, statistical quality control, and computer-controlled machines.

Our review shows that while AI offers great advantages in analyzing large datasets, spotting quality defects with cameras, and drafting routine paperwork, it has clear real-world limitations. AI platforms can generate false technical information, lack physical common sense about machine setups, operate as unexplainable black boxes, and cannot assume legal responsibility for worker safety under Canadian engineering law.

Because of these limitations, manufacturing is moving toward a **Human-in-the-Loop future**. In this setup, AI handles routine data processing and paperwork drafts, while human industrial engineers provide the practical problem-solving, common sense, shop-floor leadership, and ethical care that keep factories operating smoothly and safely.

For first-year engineering students, the message is simple: **AI will not replace industrial engineers; but industrial engineers who master core engineering concepts and use AI tools wisely will lead the manufacturing industry forward.**

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